

Hidden Layers

PREDICTIVE SALES ANALYTICS ENGINE

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WHAT WE DID IN PHASE 1

- Built a hybrid model (tabular + text embeddings)
- Used classical ML + shallow DL

PHASE 1 RESULTS

- Tabular MLP: F1 = 0.9684
- Text-only MLP: F1 = 0.5547 (weak)
- Hybrid (Late Fusion): F1 = 0.9724 (best)

MOTIVATION

- Text signal was poorly used
- PCA-20 suspected as the problem

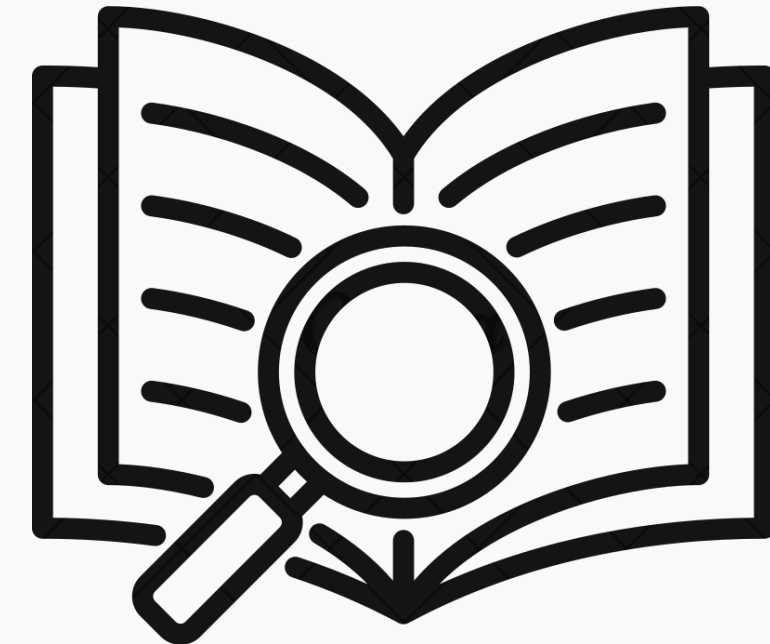


WHAT WE TRIED

- Phase 1 used MiniLM 384-dim embeddings → PCA-20
- Question: Are weak text results due to the embeddings or the PCA step?
- Switched to dataset's native 3072-dim pre-computed embeddings
- Applied the same PCA-20 reduction
- Re-trained both text-only and hybrid models

RESULTS

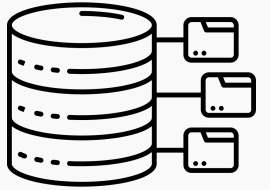
- Text-only F1: 0.5547 → 0.6215 (+6.7 pts)
- Hybrid F1: 0.9724 → 0.9687 (slight drop, -0.4 pts)
- PCA variance retained: 68.7% → 60.1% (got worse)
- Better embeddings gave worse hybrid performance



KEY FINDING

- PCA-20 is the real bottleneck — not the embedding source
- Compressing 384/3072 dims to 20 throws away most language signal
- This directly motivates Phase 2

PHASE 2 THESIS & OBJECTIVES



THESIS

- Replace PCA text with a Transformer encoder
- Add cross-modal attention between tabular and text

OBJECTIVES

- Preserve full text representation (no compression)
- Let tabular "query" the conversation
- Make predictions interpretable (show which turns matter)
- Beat Phase 1's $F1 = 0.9724$

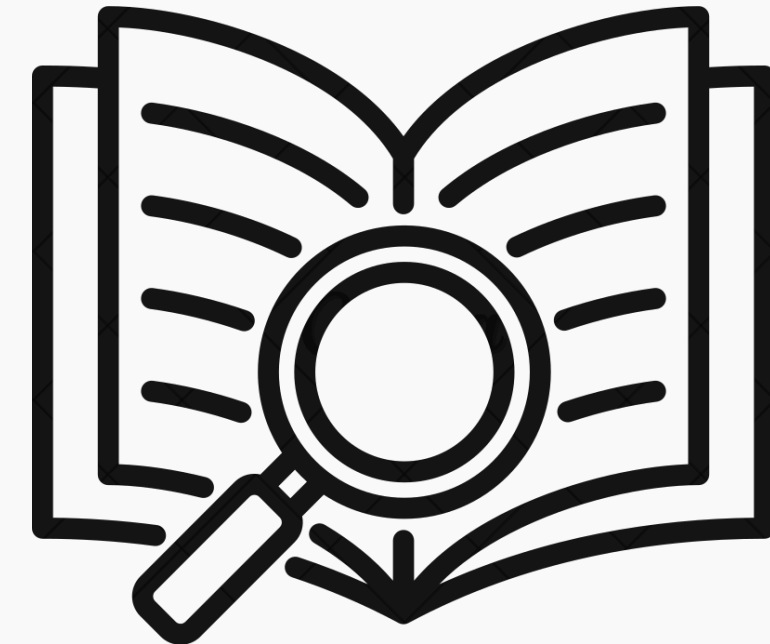


KEY PAPERS

- SalesRLAgent (2024) – RL agent for live sales prediction
- TTT – Bonnier et al. (2024) – test-time training for sales
- DistilBERT – Sanh et al. (2019) – compact BERT

INSIGHTS

- Most work uses text OR tabular, rarely both deeply fused
- Attention-based fusion under-explored in sales
- Interpretability matters for sales teams



GAP WE FILL

- Deep text + cross-modal attention + per-sample gating

REFERENCES

- SALESRLAGENT
- REVISITING MULTIMODAL TRANSFORMERS FOR TABULAR DATA WITH TEXT FIELDS

DUAL-STREAM DESIGN

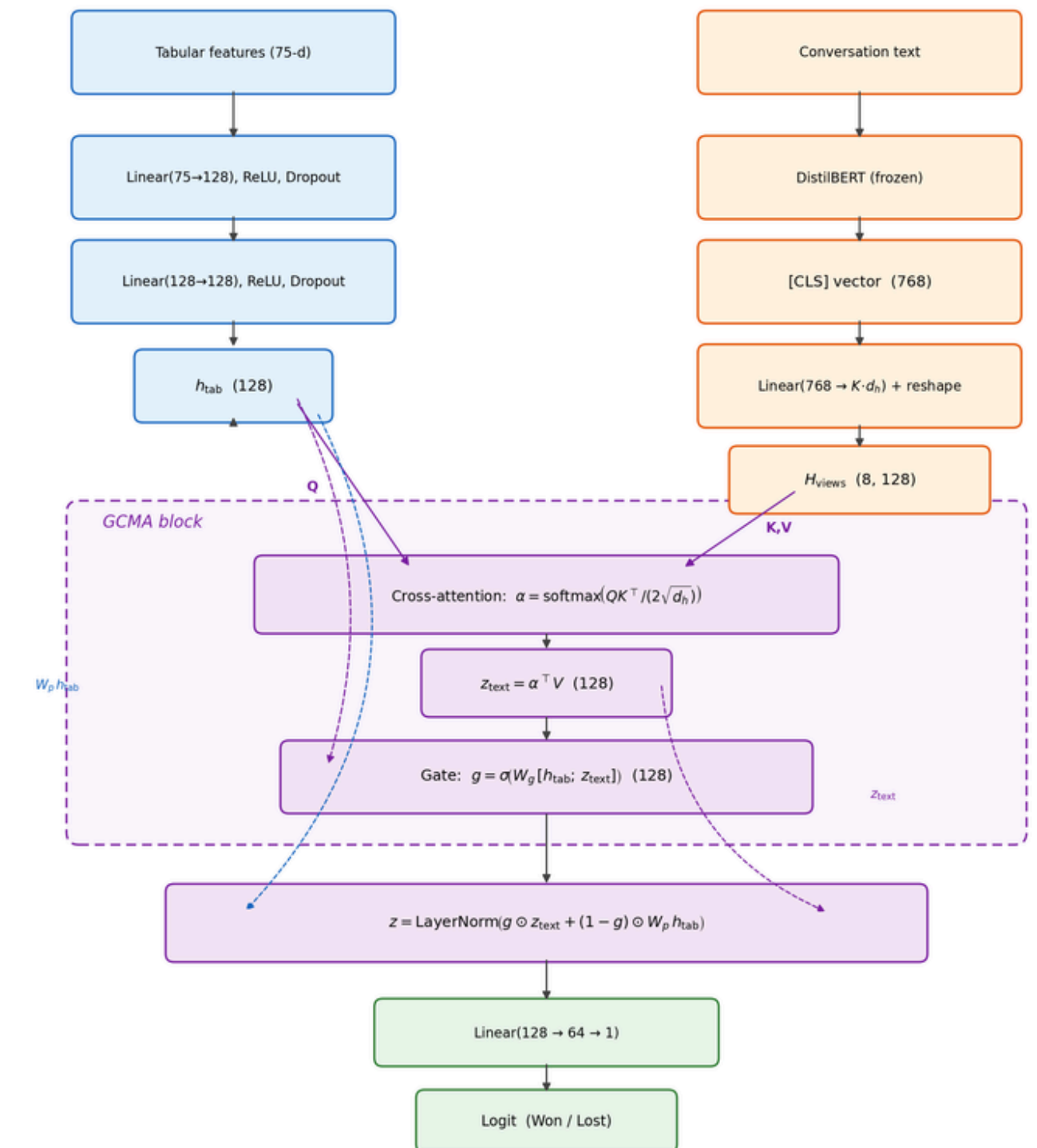
- Text branch: Frozen DistilBERT $\rightarrow$ [CLS] $\rightarrow$ 768-dim vector
- Tabular branch: MLP over 75 CRM features
- Fusion: GCMA block (attention + gate)
- Output: Final classifier $\rightarrow$ Win / Loss

HOW IT FLOWS (FOLLOW THE DIAGRAM)

- Tabular encoder creates the query (Q)
- Text encoder creates the keys & values (K, V)
- Attention selects relevant text content
- Gate blends tabular + attended text per sample
- Classifier predicts outcome

WHY FROZEN DISTILBERT

- 40% fewer params than BERT
- Retains ~97% of language understanding
- Fits 2-day training budget
- Avoids overfitting on 8K samples



GCMA: OUR NOVEL BLOCK



GATED CROSS-MODAL ATTENTION

- Q = from tabular
- K, V = from text
- Tabular asks, text answers

THE GATE

- Trust weight (0–1) per sample
- Decides how much to believe the text
- Example: small deal → trust text more; big enterprise deal → trust tabular more

WHY THIS MATTERS

- Fixes Phase 1's weak concat fusion
- Gives per-sample adaptive mixing
- Produces interpretable attention weights

ATTENTION MATH

- $z = \text{softmax}(QK^T / \sqrt{d}) \cdot v$
- Scaling by $\sqrt{d}$ keeps gradients stable

GATED FUSION

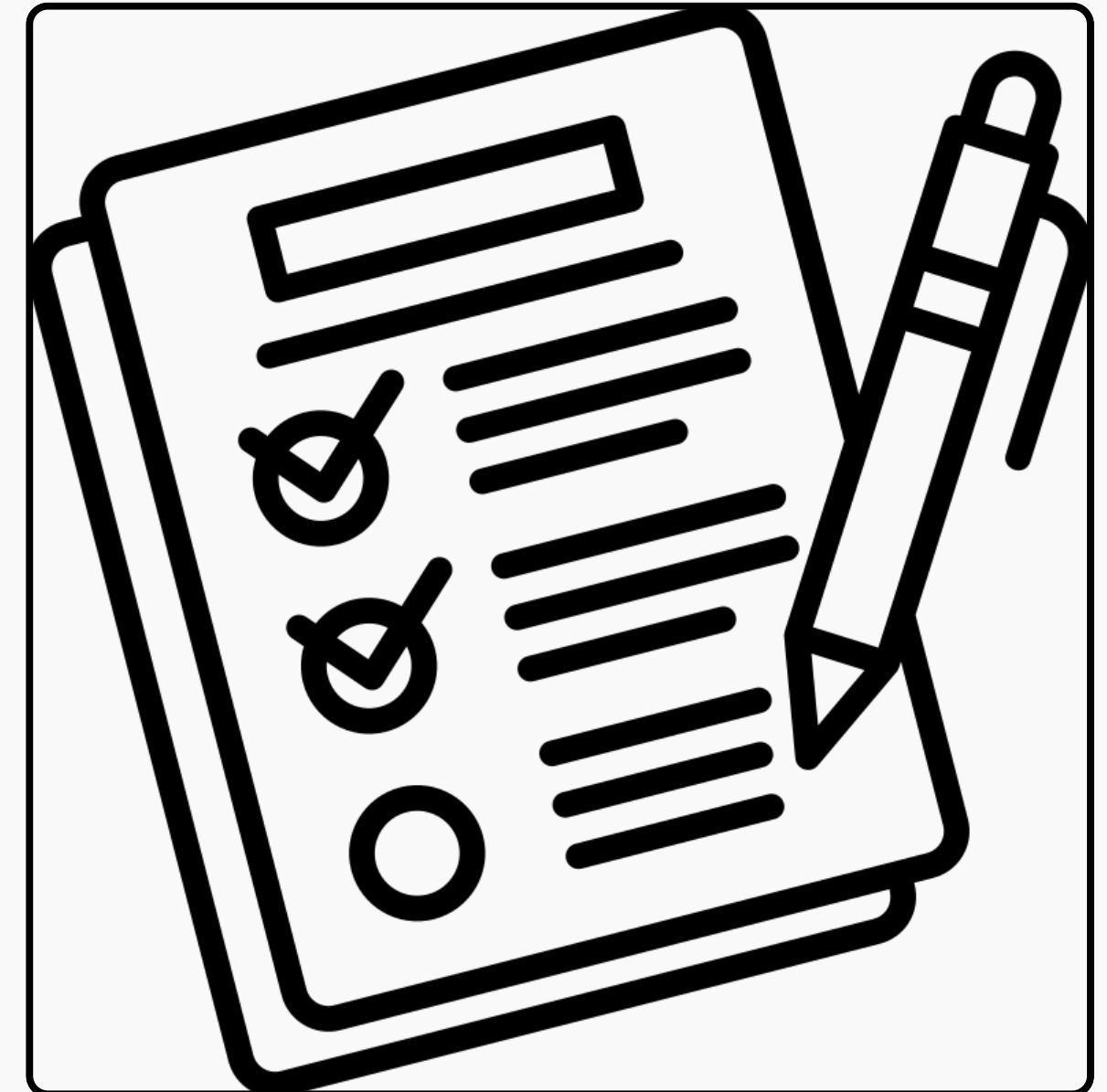
- $z_{\text{fused}} = g \odot z_{\text{attn}} + (1 - g) \odot h_{\text{tab}}$
- $g = \text{sigmoid}(\text{learned weights})$

LOSS & OPTIMIZATION

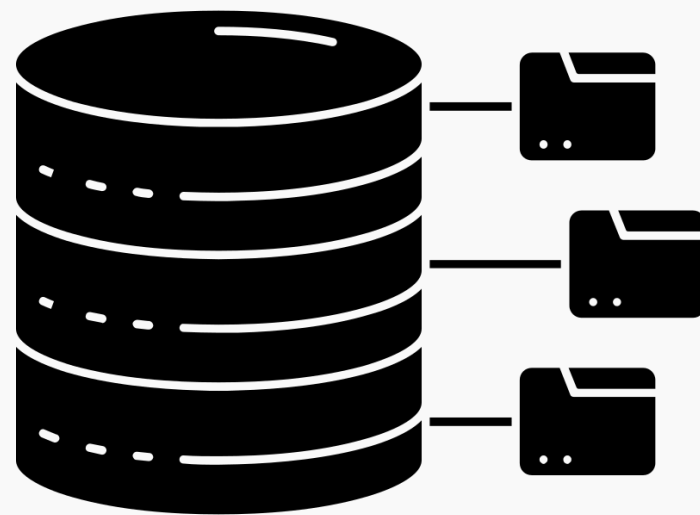
- Binary Cross-Entropy with logits
- AdamW + weight decay + LR warmup

DESIGN CHOICES

- LayerNorm (not BatchNorm) → stable with small batches
- Dropout + early stopping → avoid overfitting



DATASET, PREPROCESSING & TRAINING SETUP



DATA

- 8,000 samples (nearly balanced: 49.75% Won / 50.25% Lost)
- 70 / 15 / 15 stratified split (seed 42)

PREPROCESSING

- DistilBERT tokenizer, max_length = 512
- No-leakage splits
- Tabular features scaled + encoded

TRAINING SETUP

- Optimizer: AdamW
- Scheduler: ReduceLROnPlateau
- Early stopping (patience = 7), max 50 epochs
- Regularization: dropout + weight decay

MODEL COMPARISON (TEST F1)

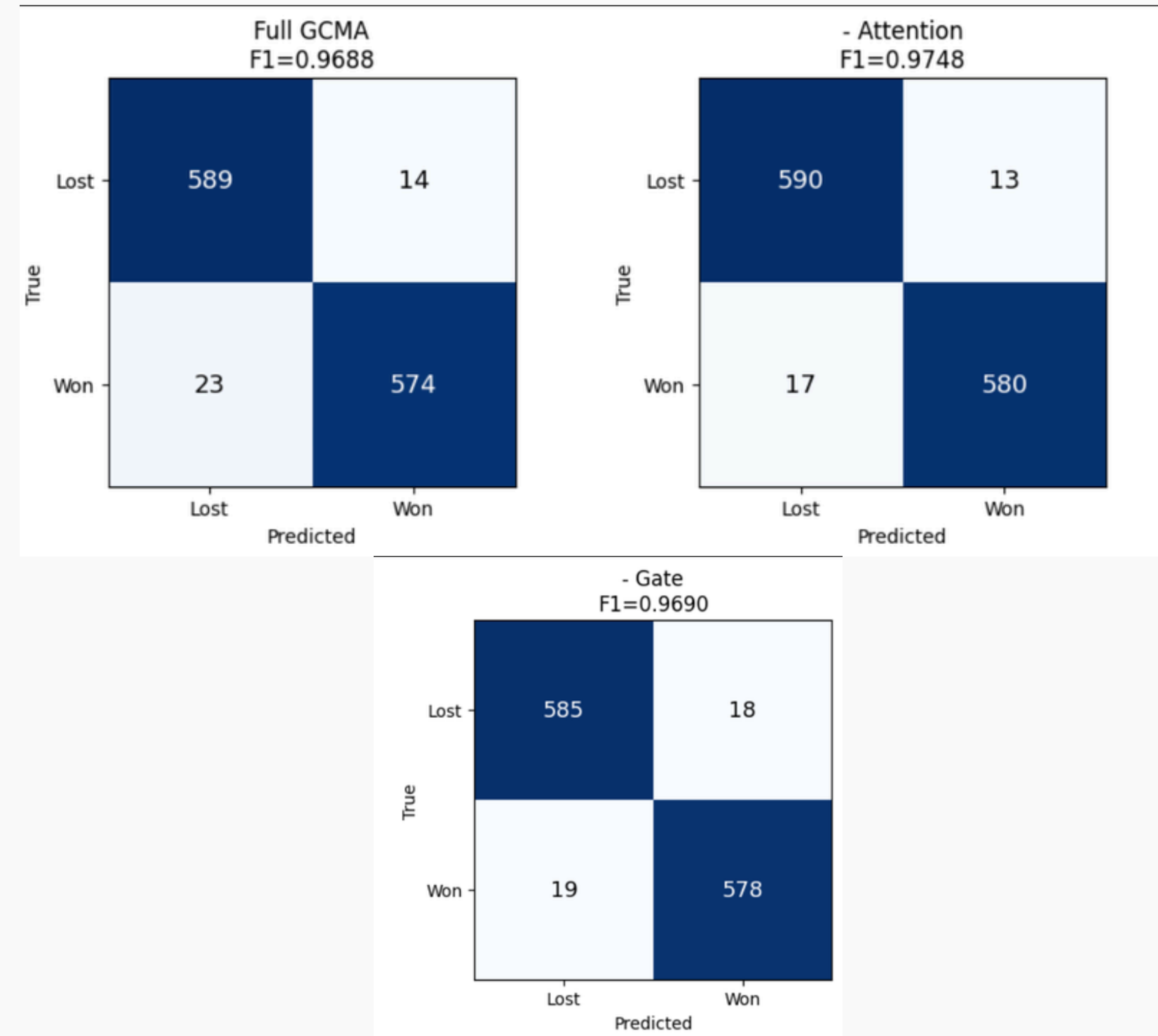
- TextMLP (DistilBERT): 0.7048 (vs 0.55 in Phase 1)
- TabularMLP: 0.9664
- GCMAFusion: 0.9688
- ConcatFusion: 0.9749 (best)

ABLATION HIGHLIGHTS

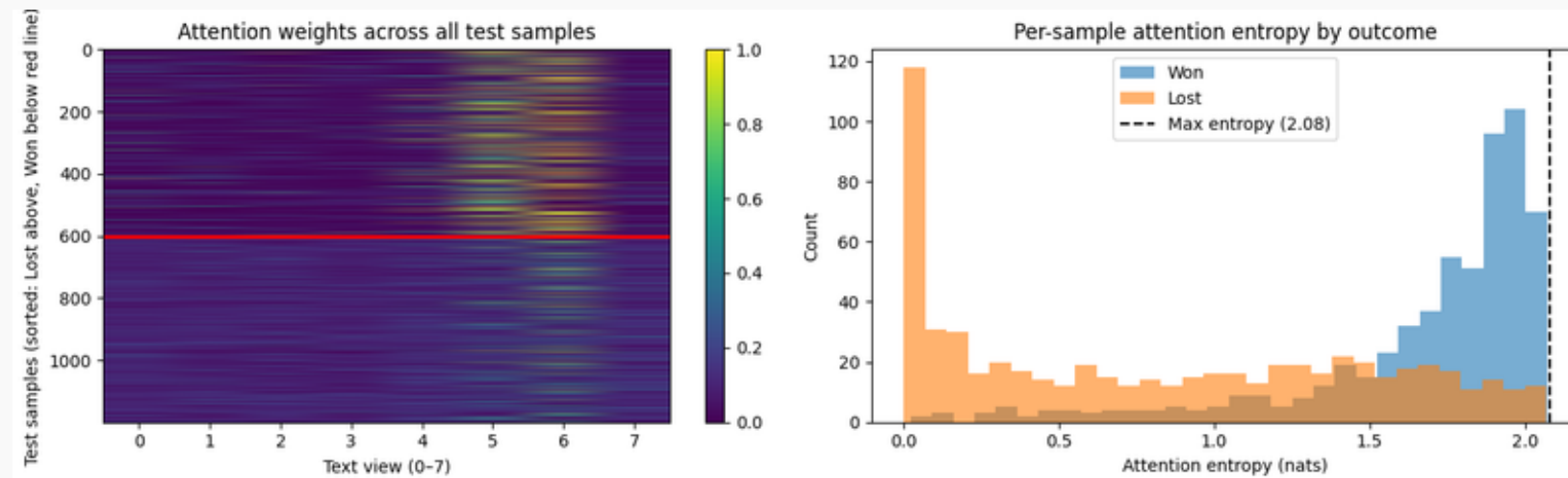
- Attention: 0.9748
- Gate: 0.9690 (gate helps slightly)
- MixUp: 0.9738
- SSL pretrain: 0.9731
- Curriculum: 0.9705

KEY TAKEAWAYS

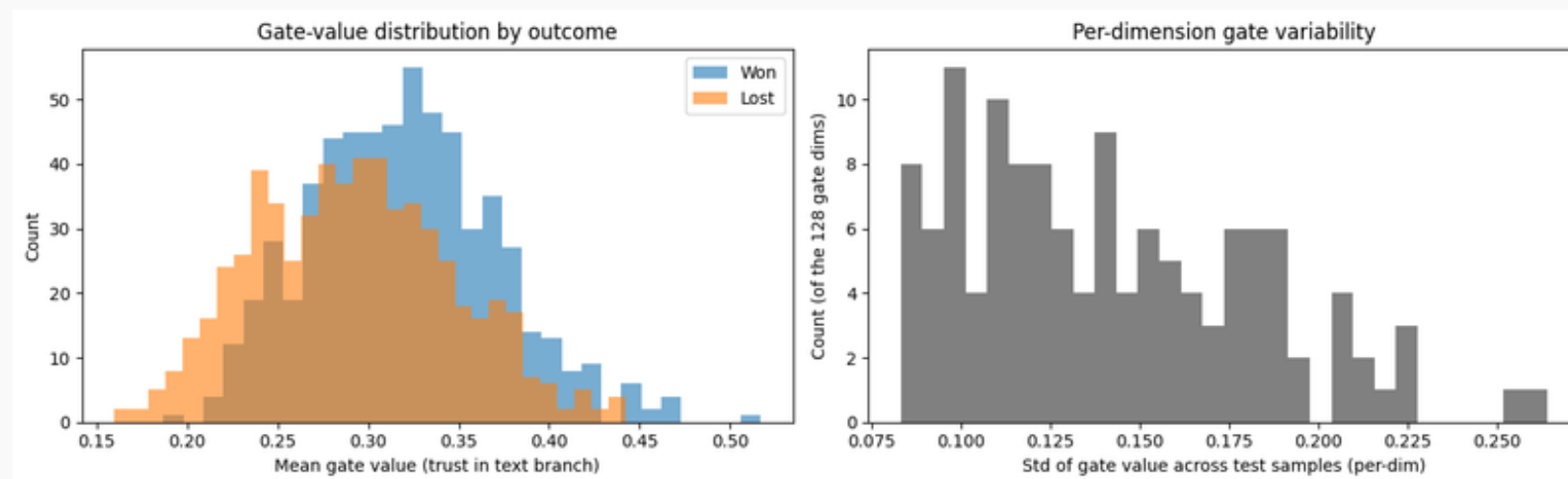
- Full 768-dim text clearly beats PCA-20
- Tabular ceiling is already close to the top
- Gains are small but consistent



ATTENTION HEATMAP



GATE DISTRIBUTION



ATTENTION ENTROPY (OVER 8 TEXT VIEWS)

- Won samples: 1.684 ± 0.408 (spread out)
- Lost samples: 0.810 ± 0.660 (concentrated)
- Class gap: 0.87 nats

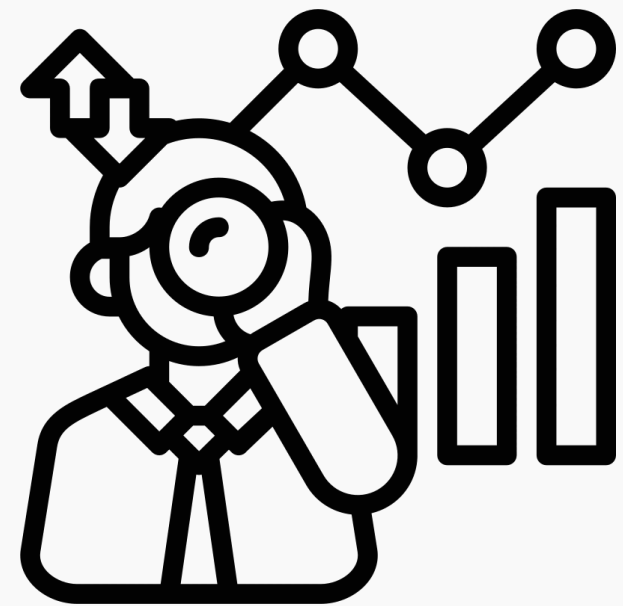
GATE BEHAVIOR

- Mean gate ≈ 0.30 – model trusts tabular more
- Per-dim std up to 0.26 $\rightarrow$ gate is not collapsed
- Won deals trust text slightly more than Lost deals

WHY THIS MATTERS

- Attention is class-discriminative, even when not raising F1
- Won deals $\rightarrow$ many turns contribute
- Lost deals $\rightarrow$ one or two turns dominate
- Gives sales teams an explainable "why"

LIMITATIONS, PHASE 3 ROADMAP



LIMITATIONS

- Frozen encoder → no task-specific text tuning
- 512-token truncation loses long conversations
- Tabular branch saturates near ~0.97 F1 ceiling
- Single-seed results – need multi-seed confirmation

PHASE 3 ROADMAP

- Hybrid Innovation → Synergistic ML ↔ DL feedback loop
- Ablation → Diagnostic (show what breaks when parts are removed)
- Architecture Diagram → Publication-ready with tensor shapes
- Reproducibility → Turn-key (Docker + setup script + README)
- Extra Mile → Web UI + CI/CD + SHAP dashboard

THANK YOU

[GET READY FOR
PHASE-3]